Data Appendix

A1. Data Sources

The dataset used in this project is the Galaxy10 DECaLS Dataset, a labeled image dataset designed for galaxy morphology classification. The dataset is publicly available through the astroNN project and contains galaxy images collected from the DESI Legacy Imaging Surveys. Galaxy class labels were derived from classifications provided by Galaxy Zoo, an astronomy project in which volunteers visually classify galaxies.

The dataset was selected because it provides a balanced benchmark for supervised image classification tasks involving astronomical objects.

A2. Data Collection

The Galaxy10 DECaLS dataset contains 17,736 labeled galaxy images. Each observation consists of:

- A color image of a galaxy
- A corresponding numerical class label

The images are stored in HDF5 (.h5) format and were loaded into Python using the h5py library.

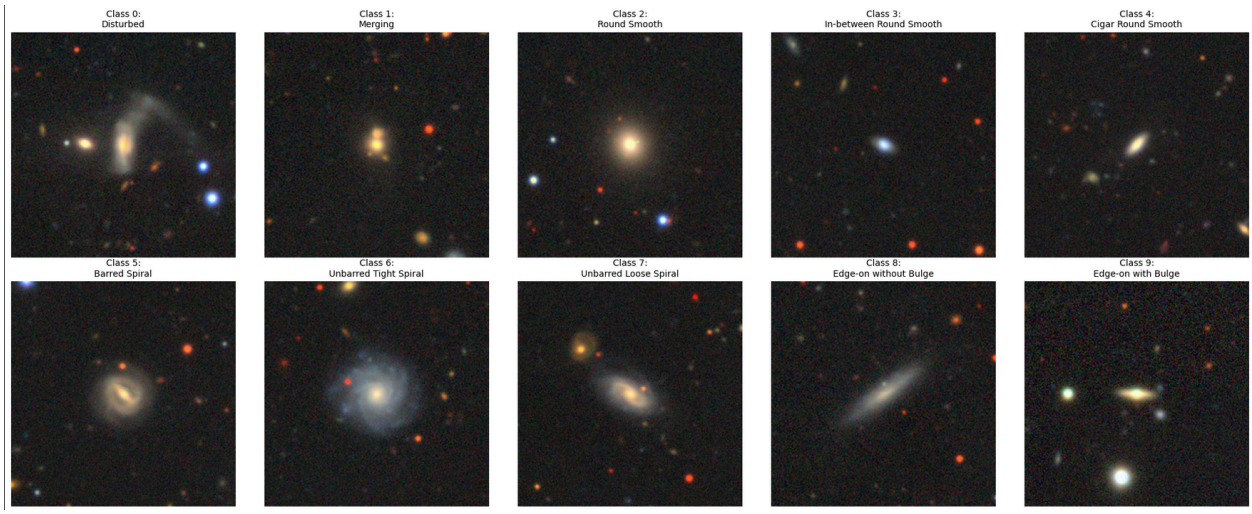
Due to the size of the file, go to this website: https://astronn.readthedocs.io/en/latest/galaxy10.html#download-galaxy10-decals and download the file manually, this is the best way to go about dealing with the data acquisition.

The ten galaxy classes included in the dataset are:

1. Disturbed
2. Merging
3. Round Smooth
4. In-between Round Smooth
5. Cigar Round Smooth
6. Barred Spiral
7. Unbarred Tight Spiral
8. Unbarred Loose Spiral
9. Edge-on without Bulge
10. Edge-on with Bulge

Each image represents a single galaxy and serves as one row of data for classification purposes.

Examples Below:



A3. Data Pre-Processing

Prior to modeling, the following preprocessing steps were applied:

1. Loaded image and label data from the HDF5 file
2. Normalized pixel values to improve model training performance
3. Applied image resizing as required by the neural network architecture
4. Performed data augmentation techniques such as:
    - Random rotations
    - Horizontal/vertical flips
5. Split the dataset into training and validation sets
6. Addressed class imbalance by including class weighing in our loss function to ensure that the model addressed the imbalance present for “Cigar Round Smooth”

These preprocessing steps were used to improve model generalization and reduce overfitting.

A4. Variable Construction

Image Variables

- image: RGB image array representing a galaxy
- pixel_values: Numerical pixel intensities used as model input

Target Variable

- label: Integer class label (0–9) corresponding to galaxy morphology type

Derived Variables

- predicted_label: Model-predicted class after training
- prediction_probability: Confidence score for each predicted class

A5. Reproducibility

All preprocessing and analysis were conducted using Python with the following libraries:

- NumPy
- Pandas
- h5py
- PyTorch
- Torchvision
- Matplotlib
- scikit-learn

The CNN model used in this project was based on a pre-trained EfficientNet-B0 architecture from Pytorch and fine-tuned on the Galaxy10 dataset.

Model interpretation was performed using Grad-CAM to visualize image regions most influential to classification decisions.

The project can be reproduced by downloading the Galaxy10 DECaLS dataset and following the preprocessing and modeling steps described above.